

45.7 Summary and Notes

45.7.1 Summary

Idea.

1. A model of the conditional mean constrains the whole conditional distribution: in a location model every quantile curve is the same curve shifted, and in a location-scale model the shape is frozen though the spread moves (Proposition 45.1.2, using Lemma 45.1.1).
2. The τ -quantile is exactly the set of minimizers of the expected check loss, for any distribution with a finite mean; the sample problem is a linear program, equivariant under scaling, shifts, reparameterization and monotone transformation of the response (Theorem 45.2.1).
3. A regression quantile interpolates p observations, and the numbers of residual signs satisfy $n_- \leq \tau n \leq n_- + n_0$ (Proposition 45.2.2).
4. Knight's identity splits the increment of the check loss into a linear term and a controllable remainder (Lemma 45.3.1); with the convexity lemma it gives asymptotic normality with a sandwich covariance built from the error densities at zero, and in the iid case the sparsity function (Theorem 45.3.2).
5. That density can be estimated by a difference quotient of the empirical quantile function, or avoided by rank-score tests or the pairs bootstrap; on heteroscedastic data only the last two keep their nominal coverage (Example (Which interval keeps its promise)).
6. Fitted quantile curves are ordered at the average covariate vector but can cross elsewhere; sorting them can only bring them closer to the truth, in every L^p sense (Proposition 45.3.3).
7. Squared deviation in place of absolute gives expectiles: unique, increasing in τ , equal to the mean at $\tau = 1/2$, affinely equivariant, and at the quantile level $E(\mu - Y)_+ / E|Y - \mu|$, which depends on the distribution (Proposition 45.4.1).
8. Expectile regression is weighted least squares with weights determined by the signs of the residuals; a step that leaves the weights unchanged has found the exact global minimizer (Proposition 45.4.2).
9. A GAMLSS gives every parameter of a chosen family its own structured additive predictor (Definition 45.5.1) and maximizes a penalized log-likelihood by block-wise Fisher scoring, each step a penalized weighted least squares fit; for the normal location-scale model the information is block diagonal (Theorem 45.5.1).
10. The likelihood of a flexible location-scale model is unbounded, so the smoothing parameters are what make the problem well posed and cannot be chosen by maximizing it (Example (Watching it happen)).
11. Predictive distributions are compared by proper scoring rules. The continuous ranked probability score is strictly proper, splits into a discrepancy and an irreducible floor, and is the check loss averaged over all levels; probability integral transforms are uniform under a correct model (Proposition 45.6.1).
12. On Engel's budgets the quantile model and the location-scale GAMLSS beat the homoscedastic mean model by about three paired standard errors of the score, and are indistinguishable from each other (Example (Mean model, quantile model and

GAMLSS on Engel's budgets)).

45.7.2 Notes and sources

Quantile regression. The subject begins with Koenker and Bassett (1978), which introduced the check loss, the linear programming formulation, the counting properties and the asymptotics under identically distributed errors, and with Koenker and Bassett (1982), which brought Engel's budget data into the literature as a test bed for heteroscedasticity. Koenker (2005) is the standard monograph and the source for everything in [Section 45.3](#) stated here without proof; Koenker and Hallock (2001) is a short introduction. The median regression ancestor is far older, as [Section 5.9](#) records.

The proof in Theorem 45.3.2 follows the convexity route of Pollard (1991), using the exact identity of Knight (1998); the convexity lemma is the one ingredient taken on trust. The sparsity estimate and its bandwidth are Hall and Sheather (1988); rank-score inference Gutenbrunner and Jurečková (1992); bootstrap validity Hahn (1995); computation Koenker and d'Orey (1987) and Portnoy and Koenker (1997). Crossing and its repairs: He (1997), Bondell, Reich and Wang (2010), and the rearrangement of Chernozhukov, Fernández-Val and Galichon (2010), whose L^p -contraction property is Proposition 45.3.3(b). The statistic $R_1(\tau)$ is Koenker and Machado (1999), nonparametric extensions start from Koenker, Ng and Portnoy (1994), and the Bayesian route through the asymmetric Laplace is Yu and Moyeed (2001), made practical by Kozumi and Kobayashi (2011).

Expectiles. Newey and Powell (1987) introduced asymmetric least squares, proved the properties collected in Proposition 45.4.1, and gave the iterated weighted least squares algorithm; Efron (1991) developed the regression percentile view and Jones (1994) the sense in which expectiles are quantiles of a related distribution. The revival came with smoothing: Schnabel and Eilers (2009), Sobotka and Kneib (2012). The risk-measurement interest follows Gneiting (2011) and Ziegel (2016).

GAMLSS. The framework is Rigby and Stasinopoulos (2005), with software in Stasinopoulos and Rigby (2007); the ancestor is the LMS method of Cole and Green (1992). The Bayesian version is Klein, Kneib, Lang and Sohn (2015) and the boosting version Mayr, Fenske, Hofner, Kneib and Schmid (2012); randomized quantile residuals are Dunn and Smyth (1996). The survey chapter of Fahrmeir, Kneib, Lang and Marx (2021) covers all three approaches and supplied this chapter's coverage checklist.

Scoring rules. The continuous ranked probability score dates to Matheson and Winkler (1976); Gneiting and Raftery (2007) is the definitive treatment of propriety, and Gneiting, Balabdaoui and Raftery (2007) formulated sharpness subject to calibration. The probability integral transform goes back to Rosenblatt (1952) and entered forecast assessment through the prequential principle of Dawid (1984). Comparing predictions by a paired test of their score difference is Diebold and Mariano (1995).

Data. Engel's 1857 budget survey is public domain and distributed with statsmodels. It appeared in chapters 20 to 23 as an example of a difficulty; here it is the subject, and nothing in this chapter is a claim about nineteenth-century Belgium.

45.7.3 What the ladder reached, and what lies beyond it

This book began with a projection onto a subspace and ends with an estimated conditional distribution. Each part changed one thing. Parts I to III built the classical model and its exact theory; Part IV let the design be a set of factors, so that the model matrix is rank deficient by construction; Part V audited every assumption in turn and asked what survives each departure; Part VI gave up unbiasedness; Part VII dropped $\text{Cov}(\boldsymbol{\epsilon}) = \sigma^2 \mathbf{I}$; Part VIII dropped normality; Part IX dropped the linear predictor; and Part X dropped the decision to model only the mean.

What holds the sequence together is that almost nothing was thrown away at each step. The normal equations became estimating equations; the projection became a smoother matrix and then a weighted smoother matrix; the residual sum of squares became a deviance and then a penalized log-likelihood; the trace of the hat matrix became effective degrees of freedom, and it is still a trace. A reader who has followed the argument to here has not learned forty-five methods but one method, restated.

Three directions are left untouched, and it is more useful to name them than to gesture at them.

Prediction without a model. Random forests, gradient boosting and neural networks predict better than any model in this book on many problems, and they can be made distributional: quantile regression forests (Meinshausen 2006) return a conditional distribution rather than a conditional mean. What they do not return is a coefficient with an interpretation. Conformal prediction (Vovk, Gammerman and Shafer 2005) is the counterweight, wrapping any predictor in a prediction set with finite-sample coverage under exchangeability alone — a stronger guarantee than anything proved here, and a weaker statement, because it says nothing about the mechanism.

Causal inference. Chapter 25 went as far as a regression book can. The modern subject — instrumental variables beyond Definition 24.4.1, difference-in-differences, regression discontinuity, synthetic controls, the semiparametric theory behind doubly robust estimation — is a separate discipline with its own books, linked to this one by the Frisch–Waugh–Lovell theorem (Theorem 6.6.2), the engine inside double machine learning.

Data that are not vectors, and data that are not complete. Functional data analysis (Ramsay and Silverman 2005) rebuilds the linear model when the response or the covariate is a curve, with the penalized splines of Chapter 43 as its tool and the same trace as its degrees of freedom. Conditional transformation models (Hothorn, Kneib and Bühlmann 2014) learn the map to a reference law, a fourth route alongside the three of Section 45.1. Regression for event times has its own century of theory beginning with Cox (1972); and censored quantile regression (Powell 1986) belongs here too, its point being that quantiles are equivariant under the censoring map $y \mapsto \max(y, c)$ by Theorem 45.2.1(d) while means are not.

None of these is promised in a later chapter, because there is no later chapter. What this book offers is the apparatus with which to read them.

45.7.4 References

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